Microstructural Influence on Material Degradation in a Rotating Detonation Engine

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Rotating detonation engines (RDEs) impose a thermomechanical loading environment on combustor wall materials that is both extreme in magnitude and strongly nonuniform along the annulus, yet the influence of microstructure on degradation behavior under detonation exposure has not been systematically studied. This work presents the first comparative investigation of how additive manufacturing (AM)-induced microstructures govern the surface evolution and material loss mechanisms of copper RDE inner bodies. Four microstructurally distinct copper inner bodies will be tested in UC Irvine's 2-inch Mini-RDE under identical methane-oxygen detonation conditions: a cast C101 reference (equiaxed grains), a laser powder bed fusion (LPBF) build (columnar grains aligned with the build direction), and two cold spray additively manufactured (CSAM) builds with splat-boundary anisotropy oriented either transverse to the bore axis (axial spray) or parallel to the bore axis (radial spray). Preliminary results on the as-cast C101 inner body, characterized over cumulative fire times from 0 seconds to 120 seconds using Keyence digital diameter metrology, Olympus laser confocal profilometry, digital microscopy, and SEM/EDS, reveal a degradation mechanism dominated by a combination of localized melting and vaporization. SEM/EDS images taken in regions of greatest material loss show islands of metallic copper sitting above and fully encapsulated by surface oxide—evidence of fluid copper that oxidized and resolidified after detonation exposure. The full manuscript will extend these methods to the LPBF and two CSAM inner bodies, correlating material degradation mechanisms with the underlying microstructure of each build. To the authors' knowledge, this will provide the first dataset linking AM-induced microstructure to RDE inner body degradation and will begin to identify which material properties govern survivability in an RDE environment.

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I. Introduction

Rotating detonation engines (RDEs) offer pressure-gain combustion through a continuously propagating detonation wave within an annular chamber, achieving thermodynamic efficiencies that exceed those of conventional deflagration-based cycles. The wave imposes a thermomechanical loading environment that is simultaneously extreme—peak post-detonation temperatures and pressures locally exceed 3000 K and 30 bar—and strongly nonuniform along the annulus [1]. Recent work by the present authors has shown that even nominally axisymmetric copper inner bodies develop spatially distinct degradation regimes along the axis under repeated methane–oxygen detonation exposure, including substantial material loss in the downstream region and, unexpectedly, net diameter growth of approximately 400 μm in the injector-end region—not accounted for by the approximately 15 μm oxide layer alone [2].

These observations imply that wall material response is coupled to the local detonation structure and motivate the focus of this study: *how the microstructure of the inner body influences the spatial signature and dominant mechanism of material degradation.* To date, RDE inner body materials have most often been conventionally manufactured copper alloys, and while AM combustion-chamber hardware—including LPBF GRCop-42 and GRCop-84 chambers—has been tested at NASA and elsewhere, the specific influence of AM-induced microstructure on RDE inner body degradation has not been systematically isolated [3-5]. The proliferation of metal additive manufacturing (AM) for combustion-chamber components—LPBF, cold spray, directed energy deposition—now produces wall materials whose microstructures differ qualitatively from cast or wrought baselines. These AM microstructures are anisotropic by nature—particularly those produced via cold spray, a solid-state process [6]. Their interaction with the anisotropic loading of a rotating detonation wave is, to the authors' knowledge, unexplored.

The principle that microstructural orientation, not just bulk composition, governs survivability in an anisotropically-loaded high-temperature environment is well established in turbine blade engineering. Conventionally cast, equiaxed polycrystalline nickel-superalloy blades introduced in early gas turbines were limited by creep rupture initiating at grain boundaries oriented transverse to the centrifugal stress axis. The introduction of directional solidification in the late 1960s eliminated transverse grain boundaries by aligning grain growth along the principal stress direction, approximately doubling rupture life [7]. Single-crystal blades, developed in the 1980s, removed grain boundaries entirely, enabling higher operating temperatures and superalloy compositions tuned for creep performance rather than grain-boundary strengthening [8]. Each step represented a redesign of the microstructure to match the directionality of the loading environment, rather than a change of the bulk material itself.

The detonation loading in an RDE is similarly directional: the wave imposes locally anisotropic heat flux and pressure as it propagates around the annulus, and the materials now being applied to RDE inner bodies through AM are themselves inherently anisotropic. This work seeks to understand whether the alignment of the microstructural anisotropy with alignment of the wave loading governs survivability in the same way that grain orientation governs creep life in turbine blades.

By analogy, the dominant heat-spreading path in the RDE inner body is axial, away from the localized post-detonation hotspot at the offset zone and into the cooler upstream and downstream bulk, so microstructures whose internal boundaries lie transverse to that axis (CSAM axial spray) are expected to experience the greatest disruption of axial heat conduction, while those whose boundaries lie parallel to it (LPBF columnar, CSAM radial spray) preserve the conduction path. Just as eliminating transverse-to-stress grain boundaries roughly doubled turbine blade life by realigning microstructure with the creep loading direction [7], this study tests whether the analogous alignment of AM-induced anisotropy with the axial heat-spreading direction governs survivability in an RDE.

This paper presents the first comparative experimental study of microstructural influence on RDE inner body degradation. Four copper inner bodies, identical in geometry but representing four distinct microstructural states, will be tested in UC Irvine's 2-inch Mini-RDE and characterized at cumulative fire time intervals using multi-modal surface evolution analysis. In parallel, computational modeling will predict the continuum-scale anisotropic thermomechanical response of each microstructure under representative detonation loading (cyclic heat flux and pressure), enabling direct comparison between predicted stress and heat-flow distributions and the experimentally observed locations of material loss.

II. Experimental Approach

A. Inner Body Specimens

Four copper inner bodies are studied, each with the same outer geometry (matched to the UC Irvine 2-inch Mini-RDE) but distinct microstructure:

1) **As-cast C101**. Conventionally cast oxygen-free electronic copper; equiaxed grain structure as a reference.

2) **LPBF copper**. Produced at Carnegie Mellon University on an LPBF system; characteristic columnar grain morphology elongated along the build direction, with melt pool boundaries and solidification texture.

3) **CSAM copper (axial spray)**. Produced at UCI on a SPEE3D WarpSPEE3D unit by spraying along the cylinder axis. The resulting splat boundaries lie in planes *transverse* to the bore axis.

4) **CSAM copper (radial spray)**. Produced at UCI on the WarpSPEE3D by spraying radially inward onto a rotating mandrel. Splat boundaries lie in planes *parallel* to the bore axis and hoop direction.

All inner bodies are machined to identical final dimensions following deposition and polished to a 1 μm grit finish, so that geometry, surface finish, and pre-fire roughness are controlled across specimens. Pre-fire characterization—digital micrometer, digital microscopy, and laser confocal profilometry—establishes a geometric and microstructural baseline for each inner body, alongside SEM/EBSD performed on companion specimens fabricated alongside each inner body using identical process parameters.

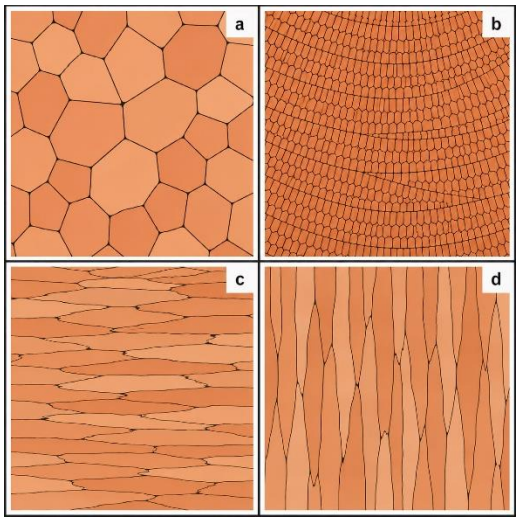


Fig. 1 Expected microstructures from (a) As-Cast, (b) LPBF, (c) Axial Cold Spray, and (d) Radial Cold Spray.

B. Hot-Fire Testing in the UCI Mini-RDE

Tests are performed in UCI's 2-inch Mini-RDE using gaseous methane and gaseous oxygen at a fixed nominal equivalence ratio and mass flow. Each inner body is fired in 2-second increments—consistent with short-duration burns typical of uncooled RDEs—with the engine allowed to cool to below 50 °C between firings. The inner body is removed for non-destructive characterization after 20 seconds of cumulative fire time, and this cycle is repeated until a total fire time of 120 seconds—at which point the inner body will be destructively characterized (SEM, EDS, EBSD). This staged-fire protocol, established in prior work on the as-cast C101 body, enables time-resolved tracking of surface evolution on a single specimen and comparison across the four microstructures.

C. Multi-Modal Surface Characterization

Between fire intervals, each inner body is characterized using a four-modality pipeline that has been developed and exercised on the as-cast C101 specimen:

1) *Keyence digital micrometer diameter measurement* along the bore axis at fine spatial resolution. The inner body is mounted on a level fixture during Keyence acquisition, so the resulting diameter profile is unaffected by tilt and serves as the geometric reference for subsequent confocal tilt correction.

2) *Laser confocal profilometry* (Evident Scientific OLS5000) for high-resolution axial surface height profiles. Tilt correction is performed by fitting the confocal raw data to the Keyence spline so the corrected profile follows the known physical bore geometry, since asymmetric material evolution (loss at one end, growth at the other) progressively tilts the part on the confocal stage and would otherwise skew the corrected profile with the very signal being measured.

3) *Digital microscopy* (Evident Scientific DSX1000) of axial strips of the inner body surface to identify visually distinct degradation regimes and locate regions of interest.

4) *SEM imaging with EDS* of the outer surface and cross-sections taken from the regions of interest, providing microstructure-resolved evidence of the dominant material loss mechanism at each axial station.

D. Bulk Thermomechanical Characterization

For each AM microstructure, bulk thermomechanical properties will be measured—both parallel and perpendicular to the bore axis—to gauge microstructural influence on bulk material behavior. Directional solidification in LPBF and various splat boundary networks in CSAM may have significant implications on the material response pertaining to the governing mechanisms for material degradation. The following properties of interest will be measured using companion specimens in the unfired state:

1) *Thermal Conductivity:* Circumferential and axial thermal conductivity using laser flash analysis.
2) *High-Temperature Strength:* Radial and axial high-temperature strength over an RDE-relevant temperature range.
3) *Creep Strength:* Constant loading at temperatures relevant to average temperatures during fire.
4) *Hardness:* Rockwell and Vickers hardness to get bulk and local microstructural hardness.
5) *Damping Ratio:* Radial and axial damping characteristics using impulse excitation and contact vibrometry.

E. Detonation Combustion Diagnostics

To assess the influence of microstructure on detonation dynamics, combustion diagnostics are recorded for each firing under matched operating conditions. High-speed imaging through the engine exit captures the rotating detonation wave structure, providing per-test measurements of frequency, wave count, and propagation velocity. Wave frequency is also confirmed with a Fourier transform of captured acoustic data. Pressure transducers in the chamber and thermocouples through the inner body will provide complementary time-resolved records of detonation strength and thermal load. These signatures are matched across microstructures to test whether wall heterogeneity perturbs the detonation behavior in measurable ways.

III. Preliminary Results: As-Cast C101 Inner Body

The as-cast C101 inner body has been fully characterized from 0 to 120 seconds of cumulative fire time. These results establish both the methodology and the degradation mechanism against which the AM microstructures will be compared.

A. Spatially Nonuniform Bore Evolution

Overlaid Keyence diameter profiles and tilt-corrected laser confocal profiles reveal that material loss along the bore is strongly spatially nonuniform. The greatest diameter *increase* is concentrated in a region downstream of the injector face that coincides with the detonation offset zone—the non-reacting injection and mixing length—reported in studies of RDE combustors [1]; the inner body profile is monotonically decreasing thereafter.

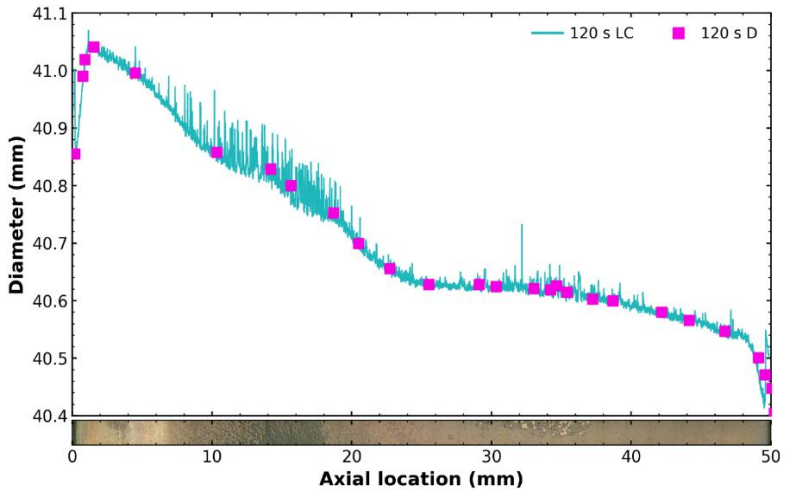


Fig. 2 Combined Keyence and tilt-corrected confocal inner body profile at 120 seconds total fire time (As-Cast C101).
Note: The unfired inner body diameter is approximately 40.6 mm.

B. Evidence for Melting and Vaporization as Dominant Loss Mechanisms

Digital micrographs of axial strips through the high-loss region show distinctive textural changes consistent with thermal degradation. SEM imaging of cross-sections taken in these regions provides direct microstructural evidence: in one representative cross-section, a distinct island of metallic copper is observed sitting *above* the local surface oxide layer and is itself *completely encapsulated by oxide*. This geometry is a signature of fluid copper that flowed during exposure, then oxidized and resolidified after the engine shutdown. Additionally, the original baseline polishing marks are visible beneath the oxide in the regime of growth near the injector side, while the copper beneath the oxide downstream by the exit reveals a surface free of marks. Together, these observations identify a combined **melting and vaporization** loss mechanism as the dominant contributor to as-cast C101 degradation.

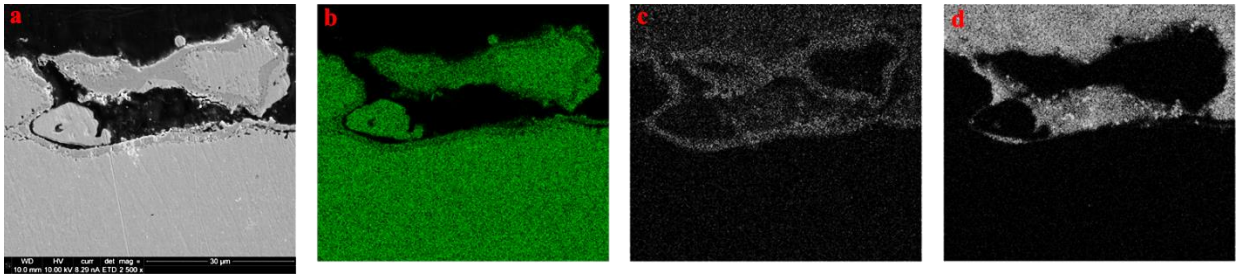


Fig. 3 SEM/EDS cross-section showing oxide-encapsulated copper, indicative of localized melting and resolidification; (a) SEM Secondary Electron Imaging, (b) Copper EDS Map, (c) Oxygen EDS Map, (d) Carbon EDS Map

IV. Expected Results

Building on the as-cast C101 baseline, we expect each AM microstructure to perturb that mechanism in a direction predictable from how its anisotropy aligns with the wave loading. The LPBF body, with columnar grains along the bore axis, preserves efficient axial heat conduction and should behave most like the cast reference. The CSAM axial-spray body, whose splat boundaries lie transverse to the bore axis, cuts across the dominant axial conduction path and should concentrate energy in the offset zone, intensifying melting. Simultaneously, the splat normals are aligned with the axial direction, exposing the splat-bonding interfaces directly to thermal cyclic stress and detonation-induced tensile loading—raising the possibility that delamination and crack growth compete with melting as a loss mechanism. The CSAM radial-spray body inverts this geometry: splat boundaries parallel to the bore axis leave axial conduction intact, but their normals align with the radial pressure pulse, so we expect a shift away from melting toward cyclic delamination of splat layers. Comparing these microstructures against the as-cast baseline may not only quantify how loading anisotropy influences material response, but also elucidate entirely new, directionally-dependent failure mechanisms not evident in the cast preliminary results.

The final manuscript will extend the methodology demonstrated on the as-cast C101 body to the LPBF, axial-spray CSAM, and radial-spray CSAM inner bodies. For each microstructure, the same staged-fire protocol (0–120 seconds in 20-second increments) and the same characterization procedure will be applied. The comparative analysis will address three questions:

1) *Does microstructure shift the dominant material degradation mechanism?* Whether the melting and vaporization observed on as-cast C101 persists, intensifies, or is replaced by a different mechanism (e.g., splat-boundary delamination and stress relief-induced cracking in CSAM, or melt pool boundary tearing in LPBF) in the AM microstructures.
2) *Does microstructural character influence detonation dynamics?* Whether the variations in interface density, dislocation density, porosity, and directional alignment of microstructural features that distinguish the four AM routes measurably alter wave count, propagation velocity, or detonation stability—particularly through their effect on the wall's thermal response and on its elastic damping of detonation-induced acoustic waves.
3) *Does microstructure influence cumulative survivability?* Whether mass loss, inner body geometry distortion, and rate of evolution with cumulative fire time depend measurably on AM route—including through the role of elastic damping in mitigating cyclic stress accumulation.

To the authors' knowledge, these results will constitute the first reported comparison of microstructurally-distinct AM inner bodies in a rotating detonation environment, and the first direct evidence linking specific AM microstructural features to RDE degradation behavior. The findings will inform both the design of more survivable RDE wall materials and the development of next-generation RDE models that incorporate wall-state evolution.

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